

02LQD Specification and Simulation of Digital Systems

2018/09/07

1. Combinational circuit design (5 points)

A combinational network with 4 inputs and 4 outputs is defined by the following truth table:

input	output
0000	0011
0001	0100
0010	0101
0011	0110
0100	0111
0101	1000
0110	1001
0111	1010
1000	1011
1001	1100
1010	1101
1011	1110
1100	1111
1101	0000
1110	0001
1111	0010

Design in VHDL a transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM-D design (25 points)

The binomial coefficient is recursively defined as follows:

$$\binom{n}{k} = n! / (k! * (n - k)!)$$

The terminal cases are:

$$\binom{n}{n} = 1 \quad \binom{n}{0} = 1$$

Iteratively it is computed as

$$\binom{n}{k} = n (n - 1) \dots (n - k + 1) / k!$$

Design a FSM-D to **iteratively** compute binomial coefficients. Inputs n and k ($n \geq k$) are pure binary numbers on 3 bits. It's up to the candidate to define the size of the `data_out` output. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**10 points**)
2. Design a FSM-D (**15 points**):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram
 - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Monday September 10 1pm

- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.